

A certified upper bound of 17 for balanced rainbow-free colorings of the four-element cube

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Abstract

Let C_4^3 be the hypergraph on $\{0, 1, 2, 3\}^3$ whose edges are its four-point geometric lines. We prove that a surjective coloring with color-class sizes differing by at most one, and with no rainbow line, uses at most 17 colors. A weighted inequality, verified on all 41,664 triples and 635,376 quadruples, reduces the 18-color case to exact cover on 140 columns and 1,020 rows. A complete finite refutation tree excludes this instance. Elementary counting excludes every larger color count. The bound improves the previously announced upper bound of 18. The exact balanced upper chromatic number remains unresolved.

1 Statement and context

Write $V = \{0, 1, 2, 3\}^3$. A *geometric line* is a four-point set

$$L = \{(x_1(s), x_2(s), x_3(s)) : s = 0, 1, 2, 3\},$$

where each coordinate function is constant, s , or $3 - s$, and at least one is nonconstant. Let E be the set of distinct such lines. A surjection $c : V \rightarrow \{1, \dots, k\}$ is *balanced* if its color-class sizes differ by at most one. It is *rainbow-free* if each line contains two distinct vertices of the same color. The maximum possible k is the balanced upper chromatic number $\chi_b(C_4^3)$.

Theorem 1. *Every balanced rainbow-free coloring of C_4^3 uses at most 17 colors. Thus $\chi_b(C_4^3) \leq 17$.*

The nearest located numerical predecessor is the announcement of bounds 12 and 18 in Table 1 of Araujo-Pardo et al. [1]. Their CCCG article [2] studies the same geometric-line model and obtains exact values in larger parameter ranges. The later preprint [3], Corollary 3, gives a general bound that evaluates to 19 here, and its Problem 1 retains the exact-value question. The earlier announcement of 18 is therefore the numerical baseline for this note. The improvement is from 18 to 17, through exclusion of the equality case of a geometry-sensitive weighted inequality.

The lower bound 12 is attributed to the cited sources; a 64-vertex witness for it is not checked here. No coloring attaining 17 is supplied, and the original exact-value problem remains unresolved. The primary-source comparison, conducted on 8 September 2026, is bounded and does not establish exhaustive priority. For broad adjacent methodology, exact finite certificates in discrete geometry also appear in Zeng et al. [4]. That reference supplies no premise for Theorem 1.

There are 48 axial lines, 24 face diagonals, and four body diagonals, so $|E| = 76$. Indeed, a line with j changing coordinates has $\binom{3}{j} 4^{3-j} 2^{j-1}$ choices. The factor 2^{j-1} identifies only the two opposite parameter orientations. Two distinct lines meet in at most one vertex: two points determine a Euclidean straight line, and each allowed direction has exactly four lattice points in the cube. In particular, a vertex pair belongs to at most one edge. All these statements concern the displayed geometric lines, with no modular arithmetic.

2 A weighted inequality on small vertex sets

Put $O = \{0, 3\}$ and $I = \{1, 2\}$. Define vertex weights

$$a(v) = \begin{cases} 2, & v \in O^3 \cup I^3, \\ 1, & \text{otherwise,} \end{cases}$$

and give a line weight $\lambda(L) = 2$ if it is axial and its two fixed coordinates both belong to O or both belong to I . Give all other lines weight 1. There are 16 vertices of weight 2 and 24 lines of weight 2; consequently

$$\sum_{v \in V} a(v) = 80, \quad \sum_{L \in E} \lambda(L) = 100. \quad (1)$$

For $A \subseteq V$, let $T(A) = \{L \in E : |L \cap A| \geq 2\}$. Write $a(A) = \sum_{v \in A} a(v)$ and $w(A) = \sum_{L \in T(A)} \lambda(L)$.

Lemma 2 (Finite weighted inequality). *For every $A \subseteq V$ of size 3 or 4,*

$$w(A) \leq a(A) + 2\mathbf{1}_{\{|A|=4\}}. \quad (2)$$

Equality holds for exactly 768 triples and 252 quadruples.

Computer-assisted proof. Order the vertices lexicographically. For every unordered pair $p, q \in V$ with $|p_i - q_i| \in \{0, 3\}$ for each coordinate, form the four points

$$((3-s)p + sq)/3, \quad s = 0, 1, 2, 3.$$

Every coordinate is integral. Every geometric line has exactly one such unordered endpoint pair, so this generates precisely E . Sort each line by vertex index, then sort the lines lexicographically.

Enumerate every increasing triple and quadruple of vertex indices. For each A , intersect it directly with all 76 lines, form $T(A)$, and compute the integer slack

$$\delta(A) = a(A) + 2\mathbf{1}_{\{|A|=4\}} - w(A).$$

The complete enumeration gives Table 1. In particular no negative slack occurs. The column totals are respectively $\binom{64}{3} = 41,664$ and $\binom{64}{4} = 635,376$. This covers all subsets in the lemma, with no symmetry restriction or other pruning. The supplied standard-library Python verifier implements these operations and retains exactly the zero-slack subsets. $\square$

For any rainbow-free partition $\mathcal{A}$ of V into triples and quadruples, each line lies in at least one $T(A)$. If q is the number of quadruples, positivity of the line weights, Lemma 2, and (1) imply

$$100 \leq \sum_{A \in \mathcal{A}} w(A) \leq 80 + 2q. \quad (3)$$

3 The tight case as exact cover

Let $\mathcal{B}$ be the complete family of equality sets from Lemma 2. Regard vertices and lines as disjoint types of objects and put

$$U = V \dot{\cup} E, \quad R(A) = A \dot{\cup} T(A) \quad (A \in \mathcal{B}).$$

These are 1,020 rows on a universe of 140 columns. An *exact cover* is a selection of rows in which each column occurs exactly once.

Table 1: Complete integer slack distribution in Lemma 2.

Slack	Triples	Quadruples
0	768	252
1	2,640	1,824
2	10,656	10,236
3	15,920	44,976
4	10,272	101,040
5	1,408	189,624
6	0	175,680
7	0	96,008
8	0	15,732
9	0	4
Total	41,664	635,376

Proposition 3. *A balanced rainbow-free 18-coloring exists if and only if the rows $\{R(A) : A \in \mathcal{B}\}$ admit an exact cover of U .*

Proof. A balanced 18-coloring has eight classes of size 3 and ten of size 4. If it is rainbow-free, (3) reads $100 \leq \sum_A w(A) \leq 100$. Every local slack is nonnegative, so every class belongs to $\mathcal{B}$. Moreover, if m_L is the number of classes touching L , then $m_L \geq 1$ and $\sum_L \lambda(L)(m_L - 1) = 0$. Since every $\lambda(L) > 0$, each $m_L = 1$. The class rows therefore cover vertices and lines exactly once.

Conversely, an exact cover partitions the vertices into equality triples and quadruples, and its line columns guarantee rainbow-freeness. If it uses q quadruples and t triples, summing their local equalities gives $100 = 80 + 2q$, hence $q = 10$. Vertex counting then gives $3t + 4q = 64$, so $t = 8$. Assigning a separate color to each row produces precisely a balanced 18-coloring. $\square$

Lemma 4 (Finite exact-cover exclusion). *The full 140-column, 1,020-row instance above has no exact cover.*

Computer-assisted proof. Use columns $0, \dots, 63$ for the lexicographically ordered vertices. Use columns $64, \dots, 139$ for the lines, ordered as in Lemma 2. Order equality blocks first by size (3 before 4), then lexicographically by their increasing vertex indices. This specifies every row without color labels or a symmetry quotient.

The certificate is a preorder tree containing one byte per node. The byte specifies an uncovered column c . At that node, a row is compatible if none of its columns has yet been covered. There is one child for every compatible row containing c , in increasing row order. The child selects that row and covers its columns. A node closes only when there is no compatible row through its chosen uncovered column. If all columns are covered, an exact cover has been found and the purported refutation is rejected.

The verifier regenerates all equality rows using the complete enumeration in Lemma 2. It checks the selected column at each node, visits every compatible row, and rejects truncated data, invalid or already covered columns, and unused suffix bytes. Replaying the supplied tree consumes 842,688 bytes and checks 842,688 nodes, with 457,598 closed leaves and maximum selected-row depth 15. No exact cover is reached.

For completeness, the soundness argument is an induction on this finite tree. A closed leaf has an uncovered column that no remaining row can cover. At an internal node, any exact cover must choose exactly one of the compatible rows through c . Every such residual instance is refuted by its

child. Thus each node represents an infeasible residual instance, including the root with all 140 columns uncovered. The tree’s origin in a dancing-links search does not enter this argument. $\square$

4 Excluding every larger color count

Proof of Theorem 1. Proposition 3 and Lemma 4 exclude 18 colors. With 19 colors, balance forces twelve triples and seven quadruples. Inequality (3) would give $100 \leq 94$, excluding that case as well.

For $20 \leq k \leq 64$, write $64 = kq + r$ with $0 \leq r < k$. A balanced partition has exactly

$$B(k) = r \binom{q+1}{2} + (k-r) \binom{q}{2}$$

within-class pairs. Each of the 76 nonrainbow lines needs such a pair, and a pair belongs to at most one line. Hence $B(k) \geq 76$ is necessary. But direct simplification gives

$$B(k) = \begin{cases} 192 - 6k \leq 72, & 20 \leq k \leq 21, \\ 128 - 3k \leq 62, & 22 \leq k \leq 32, \\ 64 - k \leq 31, & 33 \leq k \leq 64. \end{cases}$$

All these cases are impossible, and surjectivity excludes $k > 64$. This excludes every $k \geq 18$ individually; no monotonicity of balanced colorability is assumed. $\square$

5 Reproducibility and limitations

The accompanying files include the full tree, equality-block list, exact-cover input, generator, and standard-library Python verifier. From the project root, a replay that preserves the original evidence is

```
python3 evidence/verify_bound17.py --self-test \
--output manuscript/replay.json
```

Python must run with assertions enabled, without `-O` or `-OO`. The verifier reconstructs all 677,040 local subsets, compares the complete equality list and exact-cover input, replays the tree, and checks the pair budget for each $k = 20, \dots, 64$. It also rejects truncated and extended certificates, an invalid column, an already covered column, and a satisfiable one-row instance. Appendix A specifies the essential tree-checking logic.

A separately implemented check regenerated lines from directions in $\{-1, 0, 1\}^3$, with the first nonzero entry positive, and enumerated all local subsets by direct intersections. Its tree replay computed row compatibility from the union of covered columns, rather than using the supplied verifier’s row-conflict table. It reproduced the same local domain and tree statistics. This provides implementation diversity, not a proof-assistant formalization or external peer review.

The two finite lemmas are explicitly computer-assisted. Their justification is the exhaustive domain and the verified finite tree, together with the elementary arguments above. Solver status messages, floating-point relaxations, and timed-out searches are not proof dependencies. The theorem supplies an upper bound only: the exact value, an attaining 17-color construction, and extensions to other cubes are not established.

Computational and AI assistance. AI assistance was used in designing the weighted argument, implementing the search and exact verifiers, checking the proof in a separate model context, and drafting this note. A fresh model context is not a statistically independent experiment. All decisive computations use exact integer operations; no proof-assistant formalization is claimed.

A Certificate checker specification

Let `rows` be the ordered equality rows regenerated from Lemma 2. The following pseudocode uses sets; the supplied implementation uses exact integer masks. The stream position is global. Every assertion is mandatory.

```
visit(covered):
    assert covered != U                # a solution rejects the tree
    assert another byte exists
    c = read_next_byte()
    assert 0 <= c < 140 and c not in covered
    choices = [r in increasing row order:
                c in rows[r] and rows[r] intersect covered is empty]
    for r in choices:
        visit(covered union rows[r])
    # no choices: uncovered c certifies a closed leaf

visit(empty set)
assert all bytes were consumed
```

In particular, the children are determined by the reconstructed instance, not declared by the certificate. A missing branch cannot be accepted merely because the stream ends. Together with the endpoint generation and full subset enumeration in Lemma 2, this gives a specification for reimplementing the proof check without trusting the search program.

References

- [1] G. Araujo-Pardo, S. Fernández-Merchant, A. Hansberg, D. Lara, A. Montejano, and D. Oliveros. Bounding the balanced upper chromatic number. In *Discrete Mathematics Days 2024*, pp. 65–70. Editorial Universidad de Alcalá, 2024. Table 1, p. 69. https://dmd2024.web.uah.es/files/Book_Proceedings_DMD%202024.pdf.
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- [4] Z. Zeng, H. Liu, K. Ratnavelu, and O. S. Huat. Exact finite certificates for the four-dimensional ternary Borsuk problem and five-dimensional ternary kissing codes. *Preprints*, 2026. <https://doi.org/10.20944/preprints202608.1658.v1>.